Generalized Linear Models

Weeks 5 and 6

Review: Case Study Physical Activity and Menstrual Pain

In a fictional observational study, 180 adults each report their menstrual pain score for one cycle, weekly physical activity, age, and whether they have endometriosis. Participants with endometriosis tend to report more pain and less physical activity. Pain scores also vary much more among participants with endometriosis. Researchers want to determine whether:

- Physical activity is associated with pain after adjusting for age and endometriosis.
- Age and endometriosis jointly contribute to explaining pain beyond physical activity.

Discuss in groups. Develop an analysis plan.

1. How would you check whether a linear model makes sense for these data?
2. Would you recommend weighted least squares? Why or why not? If so, what should the weights be, and how would you obtain them?
3. How would you tackle each research question? Specify your model(s), hypotheses, and tests.

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Modeling binary and count data

Bernoulli and Poisson distributions

Bernoulli: used to model binary data, $y \in \{0, 1\}$.

$$y \sim \text{Bernoulli}(p), \qquad f_Y(y) = p^y (1-p)^{1-y}.$$

- $E(y) = p$ and $\text{Var}(y) = p(1-p)$.
- One parameter (p) characterizes both the mean and variance.

Poisson: used to model counts, $y \in \{0, 1, 2, \ldots\}$.

$$y \sim \text{Poisson}(\lambda), \qquad f_Y(y) = \frac{e^{-\lambda}\lambda^y}{y!}.$$

- $E(y) = \lambda$ and $\text{Var}(y) = \lambda$.
- One parameter (λ) characterizes both the mean and variance.

Modeling binary data: covariate effects

- Consider the example where we are modelling the admission rate of an MBA program (p_i) based on each applicant's college GPA (x_i) (this example is American).

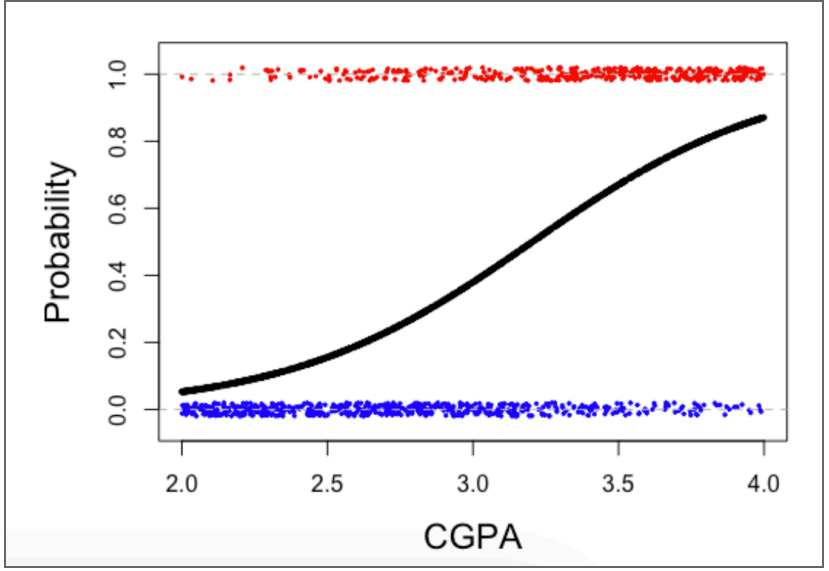
Logistic regression specifies

$$y_i \mid x_i \sim \text{Bernoulli}(p_i),$$

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + x_i\beta_1,$$

$$\implies p_i = \frac{e^{\beta_0+x_i\beta_1}}{1+e^{\beta_0+x_i\beta_1}}.$$

There is no separate additive ϵ_i term: randomness comes from the Bernoulli distribution.


Probability
0.0
0.2
0.4
0.6
0.8
1.0
2.0
2.5
3.0
3.5
4.0
CGPA

Modeling binary data: interpretation

When fitting a logistic regression, it is modelling by $y_i \sim \text{Bernoulli}(p_i)$ where

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + x_i\beta_1,$$

Interpretation of β_0

log odds of Y_i=1 when x_i=0

Interpretation of β_1

for one unit change in x_i, the log odds of Y_i=1 changes by \beta_1

Modeling binary data: divide-by-four rule

In logistic regression,

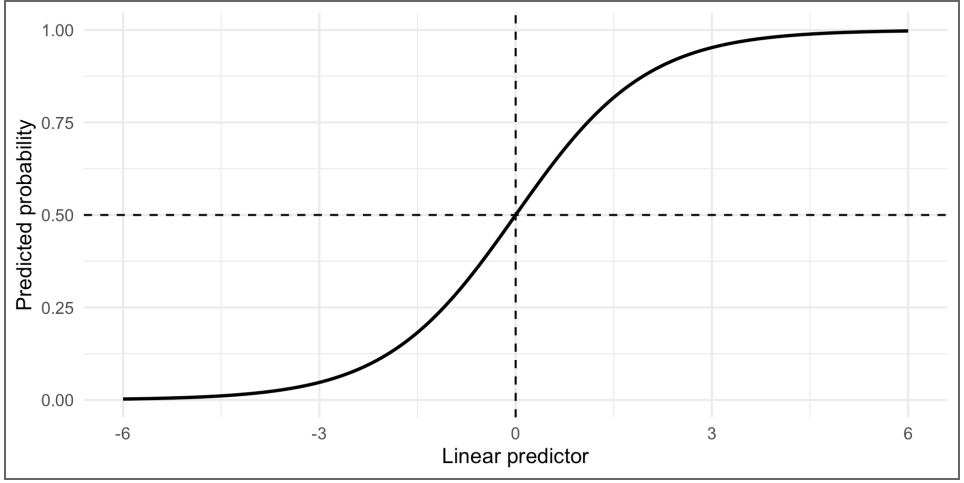
$$p = P(Y = 1 \mid X_1, \ldots, X_p).$$

Then, we can show that

$$\frac{\partial p}{\partial X_j} = \beta_j p(1 - p).$$

The quantity $p(1-p)$ is largest when $p = 0.5$ where $p(1-p) = \frac{1}{4}$.
So the **largest possible slope** is approximately $\frac{\beta_j}{4}$.

Dividing β_j by 4 gives a quick approximation to the largest possible change in predicted probability associated with a one-unit increase in X_j.


1.00
0.75
0.50
0.25
0.00
Predicted probability
-6
-3
0
3
6
Linear predictor

Modeling binary data

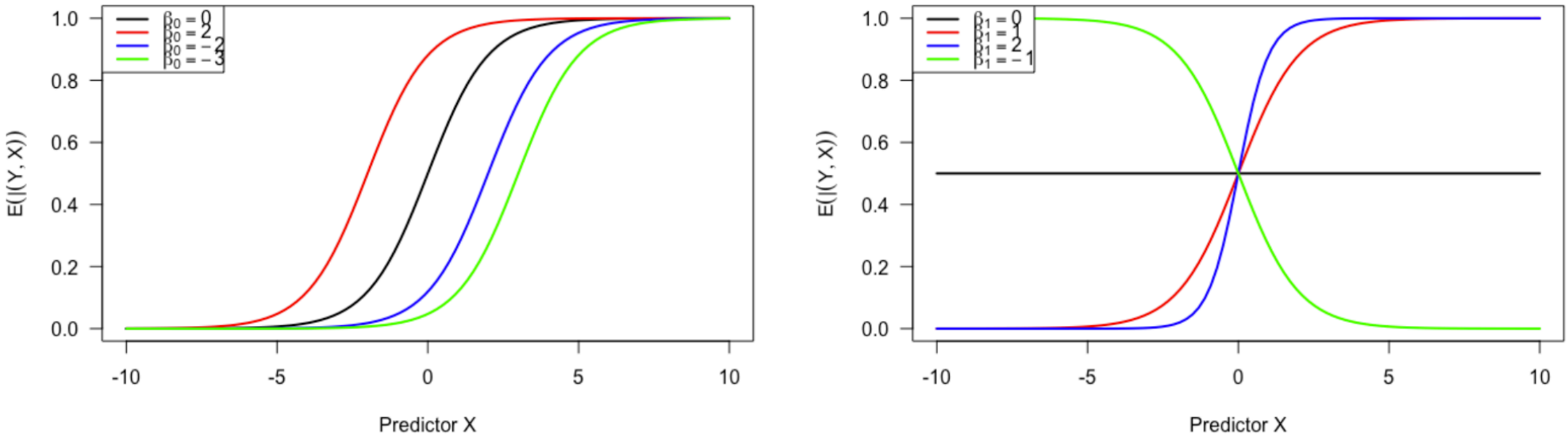


Figure. Effect of β_0 (left) and β_1 (right) on probability of $y = 1$.

Modeling binary data: different link functions

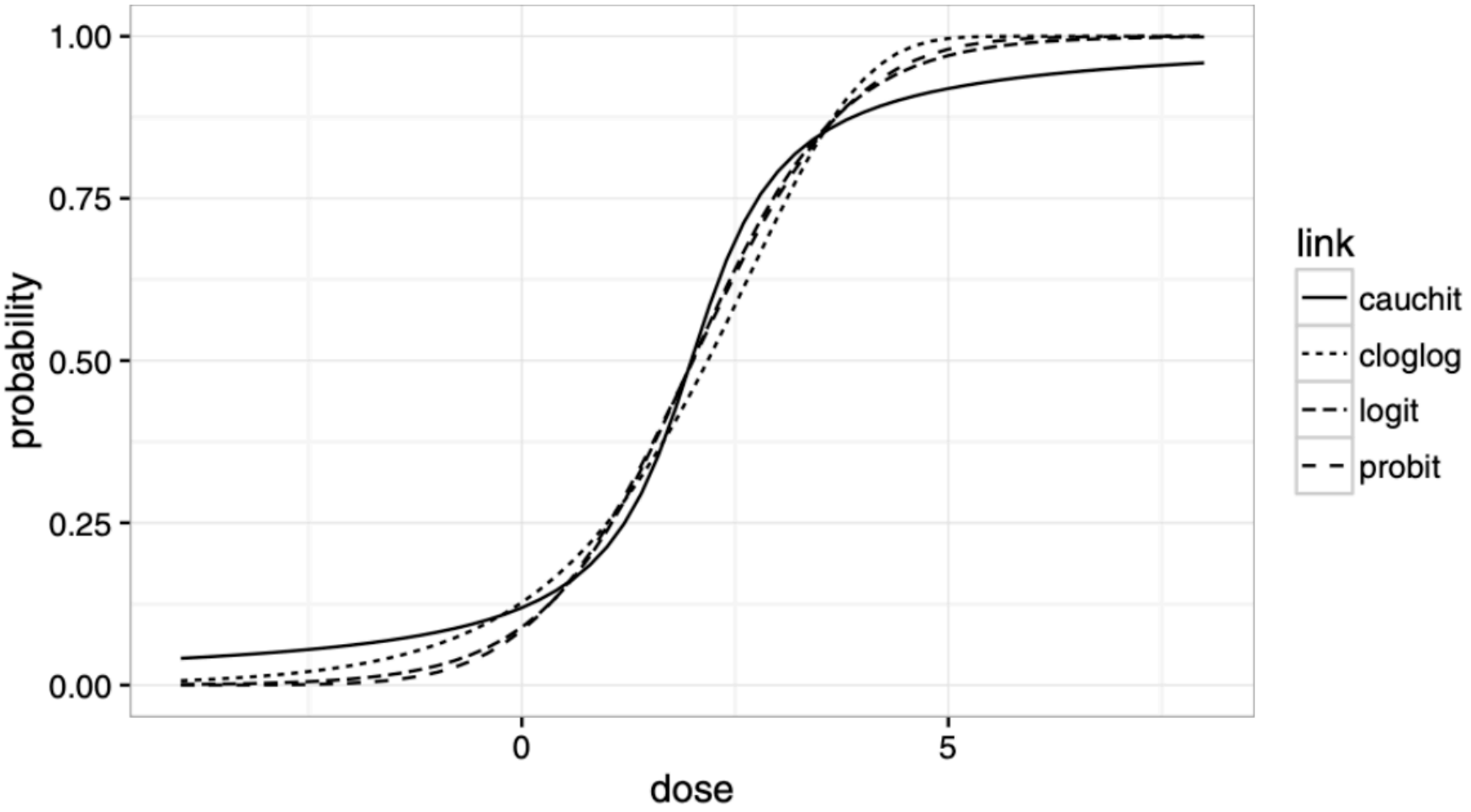


Figure 4.2 *Probit, logit, complementary log-log and cauchit compared. The data range from 0 to 4. We see that the links are similar in this range and only begin to diverge as we extrapolate.*

Modeling count data: covariate effects

Consider another example: A hospital tracks the number of emergency room visits made by each individual in 2019 (y_i). How do age, sex, and chronic health conditions affect the number of visits?

Poisson regression specifies

$$y_i \mid \mathbf{x}_i \sim \text{Poisson}(\lambda_i),$$

$$\log(\lambda_i) = \mathbf{x}_i^T \boldsymbol{\beta} \quad \Longrightarrow \quad \lambda_i = \exp(\mathbf{x}_i^T \boldsymbol{\beta}).$$

- Covariate effects are multiplicative on the expected-count scale.
- Randomness comes from the Poisson distribution, without a separate additive ϵ_i term.

Modeling count data: interpretation

When all non-intercept predictors equal zero, i.e. $x_i = 0$

$$\log(\lambda_i) = \beta_0 \quad \Longrightarrow \quad \lambda_i = e^{\beta_0}.$$

For a one-unit increase in x, holding other predictors fixed,

$$\log(\lambda_i) = \beta_0 + x_i\beta_1,$$

$$\log(\lambda_i^*) = \beta_0 + (x_i + 1)\beta_1.$$

Therefore,

$$\beta_1 = \log\left(\frac{\lambda_i^*}{\lambda_i}\right), \qquad e^{\beta_1} = \frac{\lambda_i^*}{\lambda_i}.$$

Interpretation of e^{β_1}

for a one unit increase in x_i, e^{\beta_1} is the multiplicative increase in the expected counts;

interpretation with different wording: e^{\beta_1} is the ratio of expected counts for two observations that differ by one unit in x_i, holding all other predictors fixed.

Activity: What Does the Coefficient Mean?

A researcher fits the following two models.

Model 1: Predicting Hypertension

$$\log\left(\frac{p_i}{1-p_i}\right) = -2.1 + 0.35x_i,$$

where x is age measured in **decades**.

Model 2: Predicting Number of Migraines

$$\log(\lambda_i) = 0.8 - 0.20z_i,$$

where z is hours of weekly physical activity.

Discuss:

1. Interpret 0.35 and $e^{0.35}$ in Model 1.
2. Interpret -0.20 and $e^{-0.20}$ in Model 2.

3. Why would it be incorrect to say that a one-unit increase in x increases the probability by 0.35?

Some considerations

1. Should we always use Bernoulli and Poisson models just because the response is binary or a non-negative integer?

- Despite convenience in modeling, they are often restrictive in using a single parameter to specify means and variances.
- What if $\mathrm{Var}(y) \neq E(y)(1 - E(y))$ in binary data and $\mathrm{Var}(y) \neq E(y)$ in count data? Often, there is an issue of zero-inflatedness in count modelling

2. Is it reasonable to (always) specify canonical link functions as DGP?

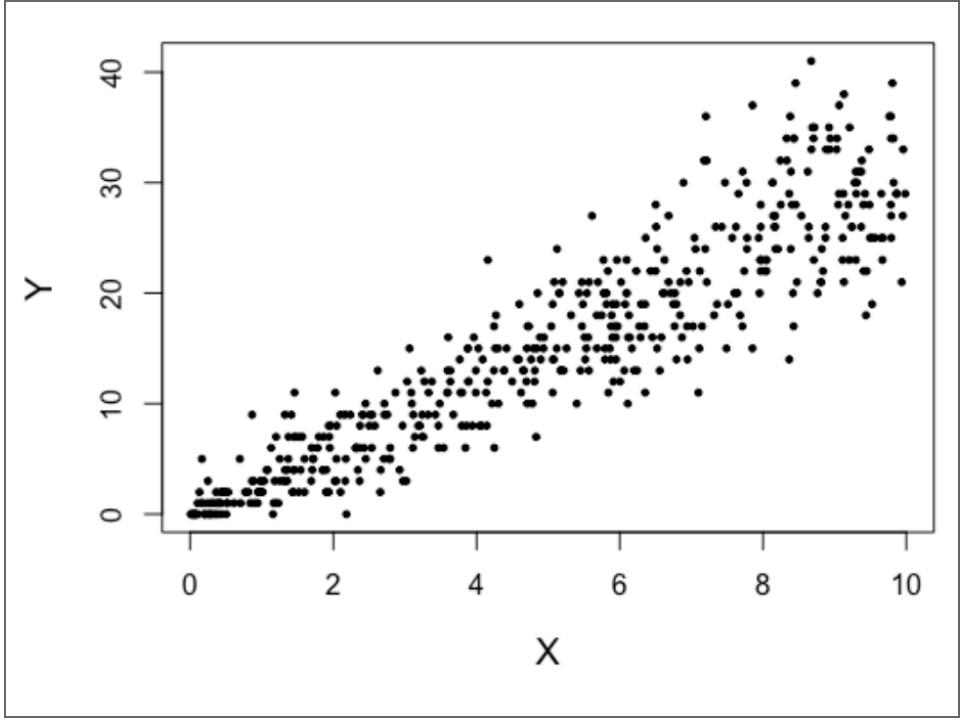
- For example, in practice, what should we choose between logit and probit link functions in logistic regression?

  In such cases, your model would be misspecified, impacting statistical inference

Example 1: can we just fit linear regression?

- Often, a linear trend is obvious when y is count data, but should we use $\exp(\boldsymbol{x}_i^\top \boldsymbol{\beta})$, (e.g. Poisson)?

- Lessons from the previous lecture: is the constant variance assumption satisfied?
- If not, what would be the impact to statistical inference when we fit linear regression?


Y
0
10
20
30
40
0
2
4
6
8
10
X

Example 2: excess zeros in count data

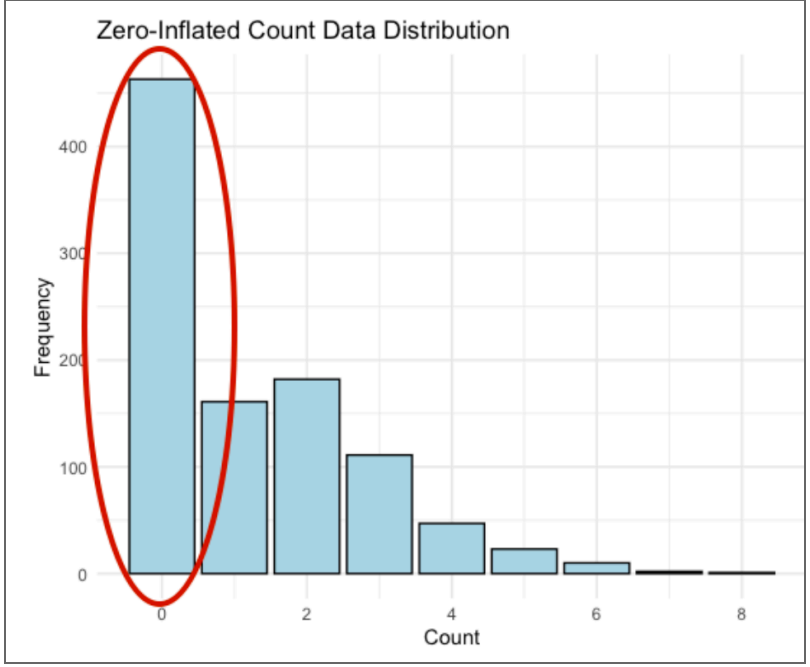
A researcher asks how many cigarettes a participant smoked in the past week.

- A non-smoker should not have consumed any
- A smoker may also report zero that week.

In such a case, a more flexible model would be in a more complicated form

$$Y_i = \begin{cases} = 0, & \text{with probability } 1-p, \\ \sim \text{Poisson}(\lambda) & \text{with probability } p. \end{cases}$$

This is an example of the Zero-Inflated Poisson (ZIP) model.


Zero-Inflated Count Data Distribution
Frequency
400
300
200
100
0
0
2
4
6
8
Count

Example 3: negative binomial counts

The negative binomial distribution can represent the number of failures before a target number of successes.

- It generalizes the geometric distribution.
- Mean and variance are specified by two parameters $\implies$ less restrictive than Poisson

$$E(Y) = \mu, \qquad \text{Var}(Y) = \mu + \frac{\mu^2}{k}.$$

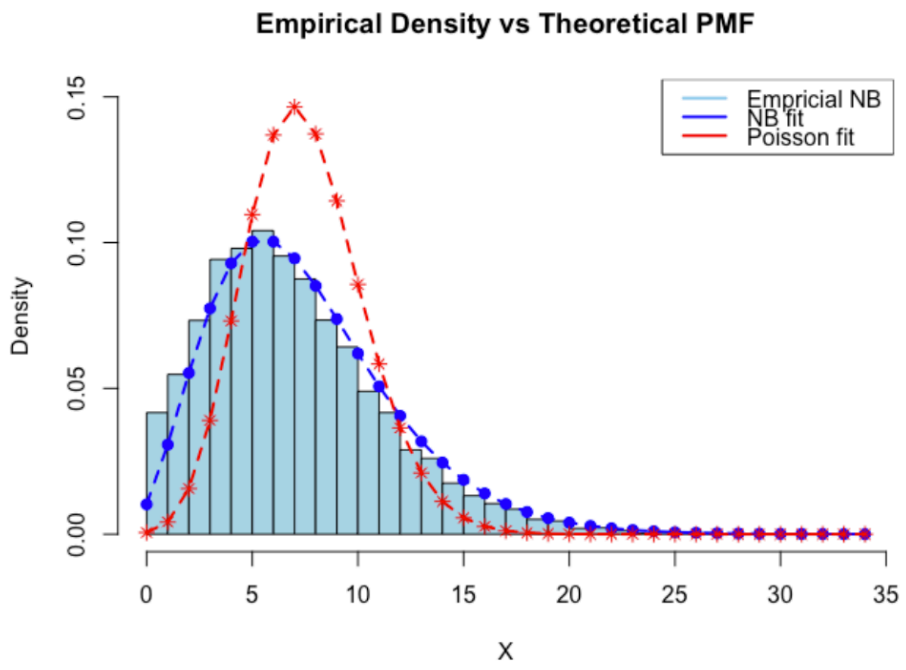
R Code | Start Over | Run Code

```r
set.seed(442)
x <- rnbinom(n = 10000, mu = 3, size = 2)
mean(x)
var(x)   # can be different from mean(x)
```

```
[1] 3.0424
```

```
[1] 7.644567
```

Poisson and negative binomial fits



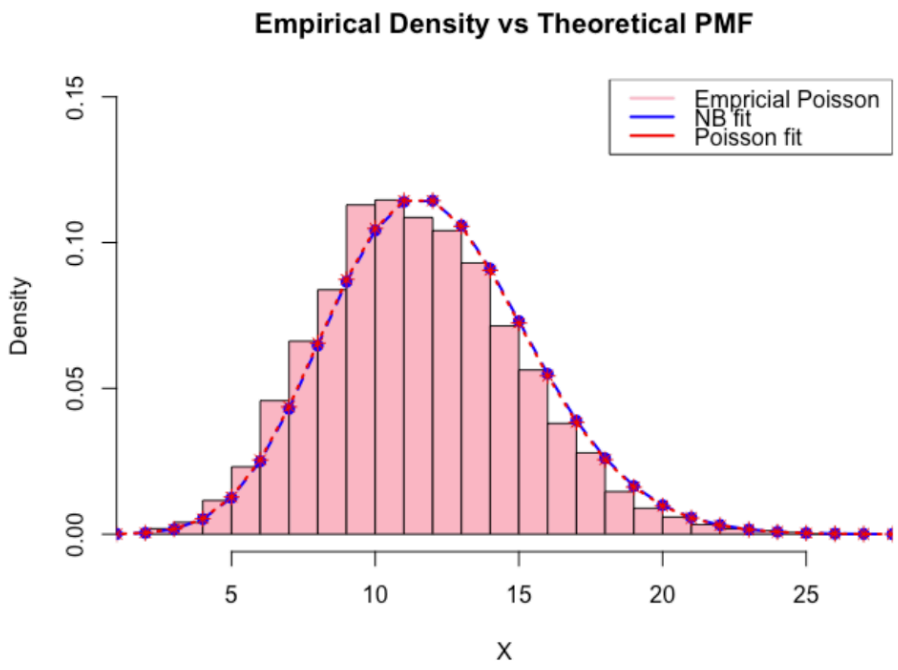


Figure. Different fits from Poisson and NB when data is generated from (i) NB (left, skyblue) and (ii) Poisson (right, pink).

Inflated Type I error?

Negative binomial data

R Code | Start Over | Run Code

```r
set.seed(0)
n <- 300
x <- rnorm(n)
p_values <- numeric(1000)
for (sim in seq_along(p_values)) {
  y <- rnbinom(n, mu = 3, size = 2)
  fit <- glm(y ~ x, family = poisson())
  p_values[sim] <- coef(summary(fit))[2,
}
mean(p_values < 0.05)
```

```
[1] 0.226
```

Zero-inflated Poisson data

R Code | Start Over | Run Code

```r
set.seed(0)
n <- 300
x <- rnorm(n)
p_values <- numeric(1000)
for (sim in seq_along(p_values)) {
  z <- rbinom(n, 1, 0.3)
  y <- ifelse(z == 0, 0, rpois(n, 3))
  fit <- glm(y ~ x, family = poisson())
  p_values[sim] <- coef(summary(fit))[2,
}
mean(p_values < 0.05)
```

```
[1] 0.285
```

Inflated Type I error?

Let's see if asymptotics help. Repeat the previous simulations with $n = 10000$. The code takes too long to run live, but when you run the code below, you will find that both Type 1 Errors are still inflated.

Negative binomial data

```r
set.seed(0)
n <- 10000
x <- rnorm(n)
p_values <- numeric(10000)
for (sim in seq_along(p_values)) {
  y <- rnbinom(n, mu = 3, size = 2)
  fit <- glm(y ~ x, family = poisson())
  p_values[sim] <- coef(summary(fit))[2, 4]
}
mean(p_values < 0.05) # 0.202 (still inflated)
```

Zero-inflated Poisson data

```r
set.seed(0)
n <- 10000
x <- rnorm(n)
p_values <- numeric(10000)
for (sim in seq_along(p_values)) {
  z <- rbinom(n, 1, 0.3)
  y <- ifelse(z == 0, 0, rpois(n, 3))
  fit <- glm(y ~ x, family = poisson())
  p_values[sim] <- coef(summary(fit))[2, 4]
}
mean(p_values < 0.05) # 0.271 (still inflated)
```

The variance assumption remains wrong as n increases. Model-based standard errors remain too small.

The next two lectures

- Learn how GLMs are estimated and how inference is performed.
- Learn remedies when the distributional assumptions behind a GLM do not hold.

Maximum likelihood and exponential family

Why maximum likelihood in GLMs?

- In the linear model with normality assumption, $\hat{\boldsymbol{\beta}}_{\text{MLE}} = \hat{\boldsymbol{\beta}}_{\text{LS}}$.
- Logistic and Poisson regression generally have no closed-form coefficient estimates.
- We estimate coefficients by maximum likelihood, using numerical optimization.
- Inference relies on MLE properties that hold asymptotically, under regularity conditions.
- These properties do not guarantee good performance in small samples.

Maximum likelihood

A quick review of the scalar parameter: θ.

"log likelihood" $\log \mathcal{L}(\theta|\text{data})$

First derivative

"score function" $S(\theta) = \dfrac{\partial \log \mathcal{L}}{\partial \theta} \longrightarrow S(\hat{\theta}_{\text{mle}}) = 0$

Second derivative

$$\frac{\partial^2 \log \mathcal{L}}{\partial \theta^2} \longrightarrow \underbrace{-E\left[\frac{\partial^2 \log \mathcal{L}}{\partial \theta^2}\right] = E[S(\theta)^2]}_{\text{"Fisher information": } \mathcal{I}(\theta)}$$

Maximum likelihood: vector parameter

A quick review of the vector parameter: $\boldsymbol{\theta}$.

"log likelihood" $\log \mathcal{L}(\boldsymbol{\theta}|\text{data})$

↓ *First derivative*

"score function" $S(\boldsymbol{\theta}) = \dfrac{\partial \log \mathcal{L}}{\partial \boldsymbol{\theta}} \longrightarrow S(\hat{\boldsymbol{\theta}}_{\text{mle}}) = \mathbf{0}$

↓ *Second derivative*

$$\frac{\partial^2 \log \mathcal{L}}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^T} \longrightarrow \underbrace{-E\left[\frac{\partial^2 \log \mathcal{L}}{\partial \boldsymbol{\theta} \delta \boldsymbol{\theta}^T}\right] = E[S(\boldsymbol{\theta})S(\boldsymbol{\theta})^T]}_{\text{"Fisher information": } \mathcal{I}(\boldsymbol{\theta})}$$

Score and information identities

The score has an expected value of 0:

$$E[S(\theta)] = 0.$$

The Fisher information is defined by the variance of the score:

$$\mathcal{I}(\theta) = \operatorname{Var}[S(\theta)] = E\left[\left(\frac{\partial \log L}{\partial \theta}\right)^2\right].$$

and, under some regularity conditions, it is also equivalent to:

$$\mathcal{I}(\theta) = -E\left[\frac{\partial^2 \log L}{\partial \theta^2}\right].$$

Why does the score have mean zero?

$$
\begin{aligned}
E[S(\theta)] &= \int \frac{\partial}{\partial \theta} \log f(x;\theta)\, f(x;\theta)\, dx \\
&= \int \frac{1}{f(x;\theta)} \frac{\partial f(x;\theta)}{\partial \theta} f(x;\theta)\, dx \\
&= \int \frac{\partial f(x;\theta)}{\partial \theta}\, dx \\
&= \frac{\partial}{\partial \theta} \int f(x;\theta)\, dx \\
&= \frac{\partial}{\partial \theta} 1 = 0.
\end{aligned}
$$

Regularity conditions allow us to interchange differentiation and integration.

Why are the information expressions equal?

$E[S'(\theta)] = -E[S(\theta)^2]$ because

$$\begin{aligned} 0 &= \frac{\partial}{\partial \theta} E[S(\theta)] \\ &= \int \frac{\partial}{\partial \theta} \{S(\theta) f(x;\theta)\}\, dx \\ &= \int S'(\theta) f(x;\theta)\, dx + \int S(\theta) \frac{\partial f(x;\theta)}{\partial \theta}\, dx \\ &= E[S'(\theta)] + E[S(\theta)^2]. \end{aligned}$$

Note that we used

$$S(\theta) = \frac{\partial}{\partial \theta} \log(f(x;\theta)) = \frac{1}{f(x;\theta)} \frac{\partial}{\partial \theta} f(x;\theta) \implies \frac{\partial}{\partial \theta} f(x;\theta) = S(\theta) f(x;\theta)$$

in the last equation above.

Properties of maximum likelihood estimators

- Asymptotic normality:

$$\hat{\boldsymbol{\theta}}_{\mathrm{MLE}} \dot{\sim} N\left(\boldsymbol{\theta}, \mathcal{I}_n(\boldsymbol{\theta})^{-1}\right).$$

- Asymptotic efficiency: $[\mathcal{I}(\theta)]^{-1}$ above achieves the Cramér–Rao lower bound asymptotically under regularity conditions, making it an efficient estimator for large samples.

- Invariance: for a one-to-one transformation h, the MLE of $h(\theta)$ is $h(\hat{\theta})$.

Maximum likelihood example: median regression

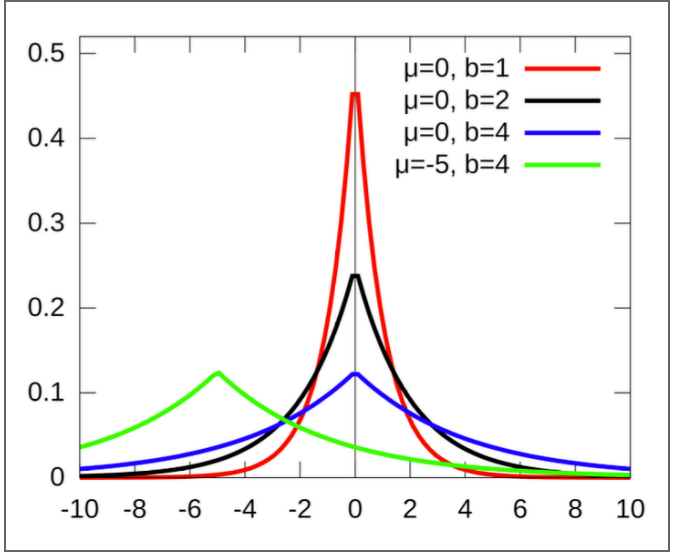
We consider a median regression, and formulate it with a Laplace distribution

$$f(y_i \mid \boldsymbol{\beta}, b) = \frac{1}{2b} \exp\left(- \frac{|y_i - \mathbf{x}_i^T \boldsymbol{\beta}|}{b} \right).$$

Then, for fixed b,

$$\ell(\boldsymbol{\beta}) = -\frac{1}{b} \sum_{i=1}^{n} |y_i - \mathbf{x}_i^T \boldsymbol{\beta}| + C.$$

Maximizing likelihood is equivalent to minimizing the sum of absolute deviations.



Under this Laplace model, median regression is more efficient than least squares asymptotically.

Exponential family

The observed values are assumed to be independent and to have come from a distribution in the exponential family.

The exponential family is characterized by a pmf or pdf of the (simplified) form:

$$f(y \mid \theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{\phi} + c(y, \phi) \right\}.$$

- θ: natural (canonical) parameter.
- ϕ: dispersion parameter.
- Normal, Bernoulli, Poisson, and Gamma distributions belong to this family.
- This framework unifies the theory behind GLMs.

Exponential family: normal distribution

We show that the normal distribution is part of the exponential family.

$$
\begin{aligned}
f(y \mid \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(y-\mu)^2}{2\sigma^2}\right] \\
&= \exp\left\{\frac{y\mu - \mu^2/2}{\sigma^2} - \frac{y^2}{2\sigma^2} - \frac{1}{2}\log(2\pi\sigma^2)\right\}.
\end{aligned}
$$

Therefore,

$$
\theta = \mu, \qquad \phi = \sigma^2, \qquad b(\theta) = \frac{\theta^2}{2},
$$

$$
c(y, \phi) = -\frac{y^2}{2\phi} - \frac{1}{2}\log(2\pi\phi).
$$

Exponential family: Bernoulli distribution

We show that the Bernoulli distribution is part of the exponential family.

$$\begin{aligned} f(y \mid \mu) &= \mu^y (1-\mu)^{1-y} \\ &= \exp\left\{ y \log\left(\frac{\mu}{1-\mu} \right) + \log(1-\mu) \right\}. \end{aligned}$$

Therefore,

$$\theta = \log\left(\frac{\mu}{1-\mu} \right), \qquad \phi = 1,$$

$$b(\theta) = \log(1+e^\theta) = -\log(1-\mu), \qquad c(y,\phi) = 0.$$

Exponential family

We have the following known properties of the exponential family that come from maximum likelihood.

1. $E[\partial \log \mathcal{L}/\partial\theta] = 0$ implies that

$$E\left[\frac{y - b'(\theta)}{\phi}\right] = 0 \implies E[y] = b'(\theta)$$

2. $E[\partial^2 \log \mathcal{L}/\partial\theta^2] = -E[(\partial \log \mathcal{L}/\partial\theta)^2]$ implies that

$$-E\left[\frac{\partial^2 \ell}{\partial \theta^2}\right] = \frac{b''(\theta)}{\phi} = E\left[\left(\frac{Y - b'(\theta)}{\phi}\right)^2\right].$$

which leads to

$$\operatorname{Var}(Y) = \phi b''(\theta).$$

Link and variance functions

A canonical link makes the linear predictor equal to the natural parameter:

$$\eta = g(\mu) = \theta.$$

- Canonical links are one possible choice.
- Bernoulli alternatives include probit and complementary log-log.

The variance function satisfies

$$\mathrm{Var}(Y) = \phi v(\mu), \qquad v(\mu) = b''(\theta).$$

Exponential family: summary

Distribution	**Canonical link** $g(\mu)$	**Mean under canonical link**	$v(\mu)$	ϕ
Normal	μ	η	1	σ^2
Bernoulli	$\log\{\mu/(1-\mu)\}$	$e^\eta/(1+e^\eta)$	$\mu(1-\mu)$	1
Poisson	$\log\mu$	e^η	μ	1
Exponential	$-1/\mu$	$-1/\eta, \eta<0$	μ^2	1

Here $\eta = \mathbf{x}_i^T\boldsymbol{\beta}$.

Study tips

Become familiar with the roles and relationships of

$$g(\cdot), \quad v(\cdot), \quad b(\cdot), \quad \phi, \quad \mu, \quad \theta, \quad \eta.$$

Review differentiation rules, especially the chain rule.

Knowledge Check

I fit two Poisson regression models to the same dataset, one using a log link and another using a square-root link. Do both models assume the same relationship between the conditional mean and variance?

yes, the links don't affect the mean-variance relationship

Two fitted Bernoulli GLMs use different link functions and give different predicted probabilities for the same individual. Will their predicted conditional variances necessarily be the same?

no, not necessarily, since the variance depends on the predicted probabilities.

Knowledge Check

Suppose a Bernoulli GLM uses a probit link instead of a logit link. Is the linear predictor still equal to the natural parameter?

no. when the linear predictor = the natural parameter, that is the definition of a canonical link. In a Bernoulli GLM, the logit link is the canonical link, so we have that property. If we change to a probit link, the linear predictor is no longer equal to the natural parameter.

Spoiler for next week: Why can we use approximately normal confidence intervals for logistic regression coefficients even though the response follows a Bernoulli distribution?

because the sampling distributions of the coefficients are asymptotically normal (property of MLE estimators).

Parameter estimation in GLMs

Review: three components of a GLM

For independent responses y_i and covariates $\mathbf{x}_i$, the data-generating process for y_i has the following specification:

1. Random component: y_i is assumed to follow a distribution that belongs to the exponential family.
2. Systematic component: given covariates $\mathbf{x}_i$, the mean of y_i can be expressed in terms of the following lienar combination of predictors

$$\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}.$$

3. Link function: relates the linear combination of predictors with the transformed mean response

$$g(\mu_i) = \eta_i, \quad \text{where} \quad \mu_i = E(y_i).$$

Review: examples of GLM specifications

Assume responses are conditionally independent.

- Logistic regression with a **logit** link:

$$y_i \sim \text{Bernoulli}\left(\frac{e^{\mathbf{x}_i^T\boldsymbol{\beta}}}{1+e^{\mathbf{x}_i^T\boldsymbol{\beta}}}\right).$$

- Logistic regression with a **probit** link, where $\Phi(\cdot)$ is a CDF function of $\mathcal{N}(0,1)$

$$y_i \sim \text{Bernoulli}\left(\Phi(\mathbf{x}_i^T\boldsymbol{\beta})\right).$$

- Poisson with a **log** link:

$$y_i \sim \text{Poisson}(e^{\mathbf{x}_i^T\boldsymbol{\beta}}).$$

- Poisson with an **identity** link (when $\mathbf{x}_i^T\boldsymbol{\beta} > 0$ is always satisfied):

$$y_i \sim \text{Poisson}(\mathbf{x}_i^T\boldsymbol{\beta})$$

Activity: Build the GLM

Suppose

$$y_i \sim \text{Bernoulli}(\mu_i), \qquad \log\left(\frac{\mu_i}{1-\mu_i}\right) = \beta_0 + \beta_1 x_i.$$

Identify

1. The random component
2. The systematic component
3. The link function
4. The variance function, $v(\mu)$
5. Which quantities vary across observations?

Question: Why might solving for $\boldsymbol{\beta}$ require an iterative algorithm?

Fitting GLMs in R

Specify the response, predictors, family, and link in `glm()`.

- linear components: what is the response, what is the covariate?
- random component and link functions could be specified as follows:

```
gaussian(link = "identity")                    ****
binomial(link = "logit")                       ****
poisson(link = "log")                          ****
Gamma(link = "inverse")
inverse.gaussian(link = "1/mu^2")
quasi(link = "identity", variance = "constant")
quasibinomial(link = "logit")
quasipoisson(link = "log")
```

How important is the choice of link?

- If we care about inference, choosing the wrong link function can be catastrophic.

- If we care only about prediction, the choice of link function does not matter much.

GLM: maximizing the log-likelihood

Derivative of log-likelihood equals to 0:

$$\log \mathcal{L} = \sum_{i=1}^{n} \underbrace{\left[\frac{y_i \theta_i - b(\theta_i)}{\phi} + c(y, \phi) \right]}_{l_i} \quad \Rightarrow \quad S_j(\beta) = \frac{\partial \log \mathcal{L}}{\partial \beta_j} = \sum_{i=1}^{n} \underbrace{\frac{\partial \eta_i}{\partial \beta_j} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial l_i}{\partial \theta_i}}_{\because \text{ chain rule}},$$

where

$$\frac{\partial \eta_i}{\partial \beta_j} = x_{ij}$$

$$\frac{\partial \mu_i}{\partial \eta_i} = \left(\frac{\partial \eta_i}{\partial \mu_i} \right)^{-1} = \left(\frac{\partial g(\mu_i)}{\partial \mu_i} \right)^{-1} = \frac{1}{g'(\mu_i)}$$

$$\frac{\partial \theta_i}{\partial \mu_i} = \left(\frac{\partial \mu_i}{\partial \theta_i} \right)^{-1} = \left(\frac{\partial b'(\theta_i)}{\partial \theta_i} \right)^{-1} = \frac{1}{b''(\theta_i)} = \frac{1}{v(\mu_i)}$$

$$\frac{\partial l_i}{\partial \theta_i} = \frac{y_i - \mu_i}{\phi}$$

GLM: a weighted least squares equation

Combining all together:

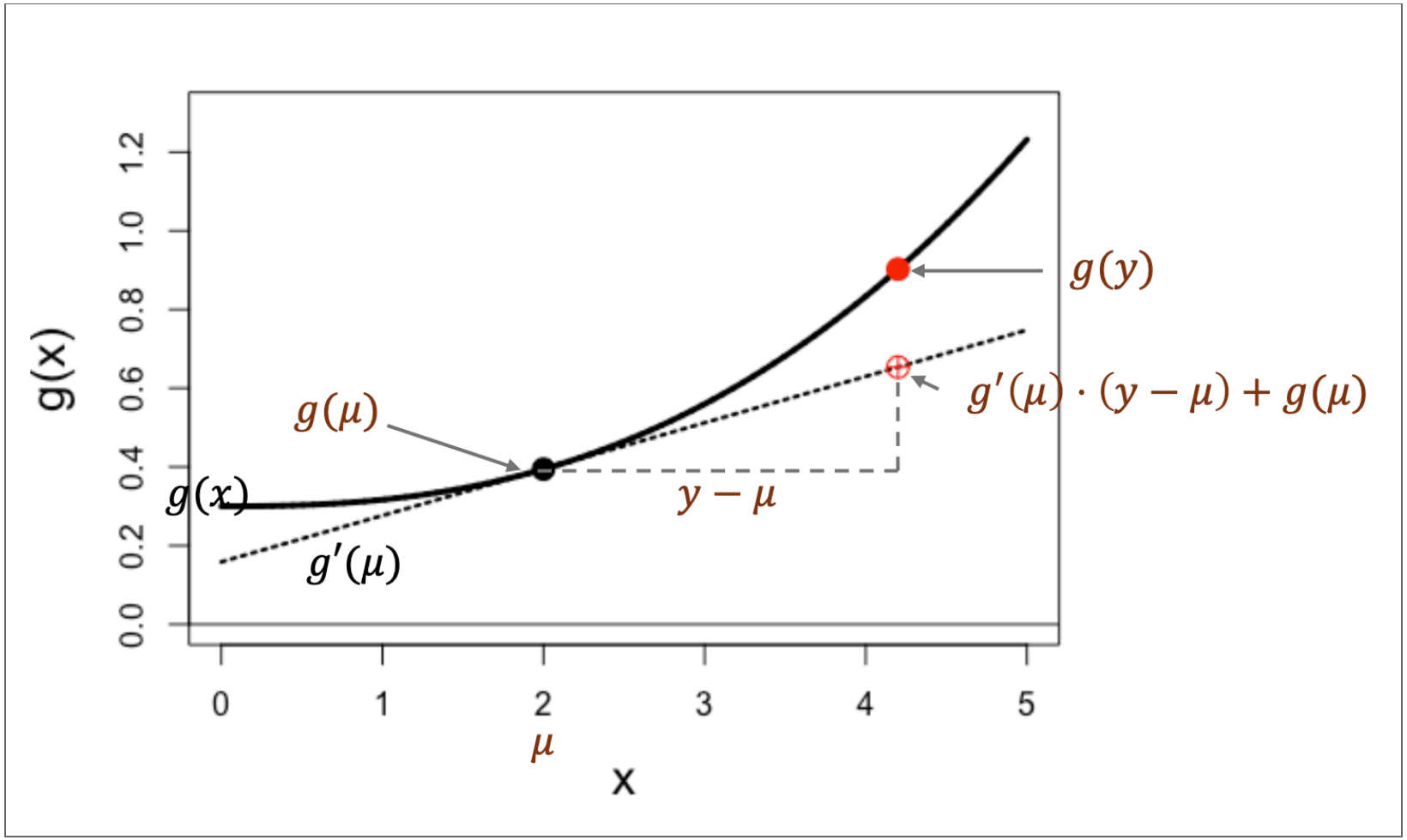
$$
\begin{aligned}
S_j(\boldsymbol{\beta}) &= \sum_{i=1}^{n} \frac{x_{ij}}{g'(\mu_i)} [\phi \cdot v(\mu_i) \cdot g'(\mu_i)]^{-1} (y_i - \mu_i) = \sum_{i=1}^{n} x_{ij} [\phi \cdot v(\mu_i) g'(\mu_i)^2]^{-1} g'(\mu_i)(y_i - \mu_i) \\
&= \sum_{i=1}^{n} x_{ij} \underbrace{[\phi \cdot v(\mu_i) g'(\mu_i)^2]^{-1}}_{\equiv w_i^{-1}} [\underbrace{g(\mu_i) + g'(\mu_i)(y_i - \mu_i)}_{\equiv z_i} - \underbrace{g(\mu_i)}_{\equiv \mathbf{x}_i^T \boldsymbol{\beta}}] \\
&= \sum_{i=1}^{n} x_{ij} w_i^{-1} (z_i - \mathbf{x}_i^T \boldsymbol{\beta})
\end{aligned}
$$

Making the above a matrix form and setting it equal to 0,

$$
\boldsymbol{X}^T \boldsymbol{W}^{-1} (\boldsymbol{z} - \boldsymbol{X}^T \boldsymbol{\beta}) = 0 \quad \text{with} \quad \boldsymbol{W} = \text{diag}\{(\phi \cdot v(\mu_i) g'(\mu_i)^2)\}
$$

which becomes a weighted least squares problem.

Understanding $g(\mu_i) + g'(\mu_i)(y_i - \mu_i)$



Note: z_i is a first-order Taylor approximation of $g(y_i)$ using μ_i.

Iteratively reweighted least squares

IRLS Algorithm

1. Initialize $\hat{\eta}_i$ (Typically, fit a linear regression).
2. Repeat the following until convergence:

$$\hat{\mu}_i \leftarrow g^{-1}(\hat{\eta}_i),$$

$$\hat{z}_i \leftarrow \hat{\eta}_i + (y_i - \hat{\mu}_i) g'(\hat{\mu}_i),$$

$$\hat{w}_i \leftarrow \phi v(\hat{\mu}_i)[g'(\hat{\mu}_i)]^2,$$

$$\hat{\boldsymbol{\beta}} \leftarrow (X^T \hat{W}^{-1} X)^{-1} X^T \hat{W}^{-1} \hat{\mathbf{z}},$$

$$\hat{\boldsymbol{\eta}} \leftarrow X\hat{\boldsymbol{\beta}}.$$

Both the working response ($\hat{z}_i$) and weights ($\hat{w}_i$) are updated each iteration.

Logistic IRLS: implementation in R

R Code | Start Over | Run Code

```r
myLogistic <- function(y, x, n_iter = 20) {
  eta <- rep(qlogis(mean(y)), length(y))
  for (iter in seq_len(n_iter)) {
    mu <- plogis(eta)
    z <- eta + (y - mu) / (mu * (1 - mu))
    w <- 1 / (mu * (1 - mu))
    fit <- lm(z ~ x, weights = 1 / w)
    eta <- fitted(fit)
  }
  mu <- plogis(eta)
  list(beta = coef(fit), wts = 1 / (mu * (1 - mu)))
}


set.seed(442)
n <- 50
x <- rnorm(n)
y <- rbinom(n, 1, plogis(0.5 + 0.2 * x))
fit_my <- myLogistic(y, x)
fit_glm <- glm(y ~ x, family = binomial())
fit_my$beta
coef(fit_glm)
```

```
(Intercept)           x
 0.48871070  0.01928739
```

```
(Intercept)           x
 0.48871070  0.01928739
```

Uncertainty Quantification

The score function is expressed by

$$\mathbf{S}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \frac{\partial \mu_i}{\partial \boldsymbol{\beta}} \frac{y_i - \mu_i}{\phi v(\mu_i)}.$$

The Fisher information matrix is

$$\begin{aligned}
\mathcal{I}(\boldsymbol{\beta}) &= \sum_{i=1}^{n} \frac{1}{\phi v(\mu_i)} \frac{\partial \mu_i}{\partial \boldsymbol{\beta}} \left(\frac{\partial \mu_i}{\partial \boldsymbol{\beta}} \right)^T \\
&= \sum_{i=1}^{n} \mathbf{x}_i \{ \phi v(\mu_i) [g'(\mu_i)]^2 \}^{-1} \mathbf{x}_i^T \\
&= X^T W^{-1} X.
\end{aligned}$$

Uncertainty Quantification

Under regularity conditions, $\hat{\boldsymbol{\beta}}$ is approximately normal:

$$\hat{\boldsymbol{\beta}} \dot{\sim} N\left(\boldsymbol{\beta}, \mathcal{I}(\boldsymbol{\beta})^{-1}\right).$$

Its asymptotic covariance is

$$\mathrm{aVar}(\hat{\boldsymbol{\beta}}) = (X^T W^{-1} X)^{-1}.$$

Since $\boldsymbol{W}$ consists of unknown parameters, we replace $\boldsymbol{\beta}$ with $\hat{\boldsymbol{\beta}}_{MLE}$ to be used in statistical inference ($\hat{\boldsymbol{W}}$)

GLM uncertainty: implementation in R

R Code | Start Over | Run Code

```r
summary(fit_glm)

vcov(fit_glm)

X = cbind(1,x)
solve(t(X)%*%diag(1/fit_my$wts)%*%X) # Manual implementation of Cov(beta.hat)
```

```
Call:
glm(formula = y ~ x, family = binomial())

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.48871    0.29159   1.676   0.0937 .
x            0.01929    0.27669   0.070   0.9444
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 66.406  on 49  degrees of freedom
Residual deviance: 66.402  on 48  degrees of freedom
AIC: 70.402

Number of Fisher Scoring iterations: 4
```

```
             (Intercept)           x
(Intercept)  0.085025788 -0.003129192
x           -0.003129192  0.076560104
```

```
                          x
   0.085025792 -0.003129192
x -0.003129192  0.076560107
```

Statistical inference in GLMs

Hypothesis Testing 1: Likelihood ratio test

$$H_0 : \underbrace{\beta_1 = \cdots = \beta_p = 0}_{\text{i.e., } \mu_i = \mu}$$

Procedure:

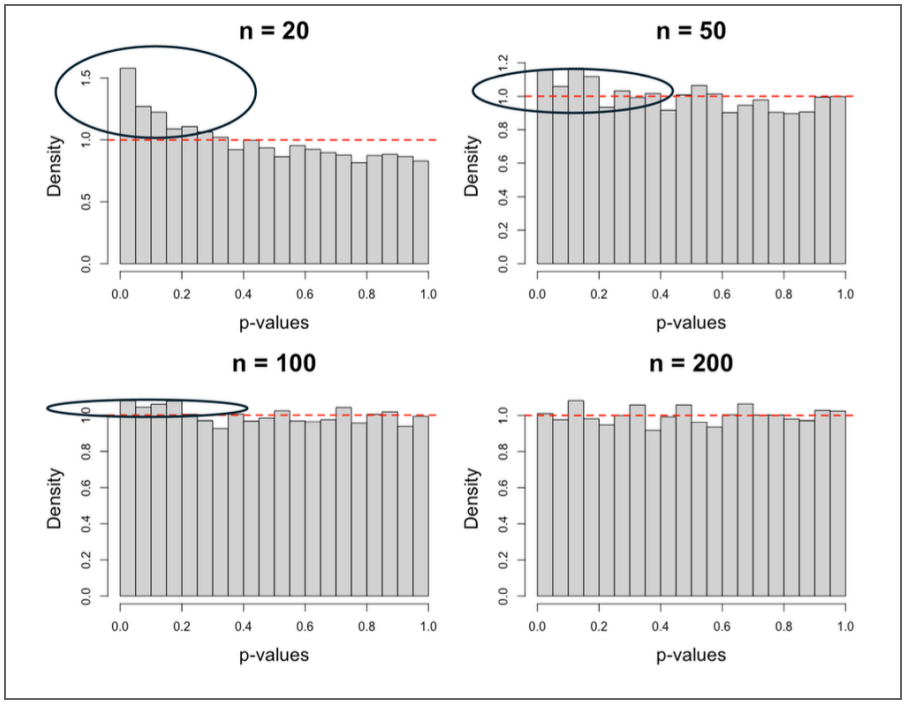
1. Fit the null model and obtain $\ell(\hat{\boldsymbol{\mu}}_{\text{null}})$.
2. Fit the full model and obtain $\ell(\hat{\boldsymbol{\mu}}_{\text{full}})$.
3. Construct the likelihood ratio statistic:

$$T_{\text{LRT}} = -2[\ell(\hat{\boldsymbol{\beta}}_{\text{null}}) - \ell(\hat{\boldsymbol{\beta}}_{\text{full}})].$$

4. $T_{\text{LRT}} \dot{\sim} \chi^2_p$ asymptotically (i.e. as $n \to \infty$) when the null hypothesis is true

Hypothesis Testing 1: Likelihood ratio test

```r
n <- 20
p_values <- numeric(10000)
for (sim in seq_along(p_values)) {
  x1 <- rnorm(n)
  x2 <- runif(n)
  y <- rbinom(n, 1, 0.3)
  fit0 <- glm(y ~ 1, family = binomial())
  fit1 <- glm(y ~ x1 + x2,
              family = binomial())
  stat <- 2 * as.numeric(
    logLik(fit1) - logLik(fit0))
  p_values[sim] <- pchisq(stat, df = 2,
                          lower.tail = FALSE)
}
```



Hypothesis Testing 2: Wald Test

$$H_0 : \underbrace{\beta_1 = \cdots = \beta_p = 0}_{\text{i.e., } \mu_i = \mu}$$

Procedure:

1. Estimate $\hat{\beta}_{1:p}$ using MLE
2. Obtain $\widehat{Var}(\hat{\beta}_{1:p})$, denoted by $\hat{\Sigma}_{1:p}$.
3. Construct the Wald statistic

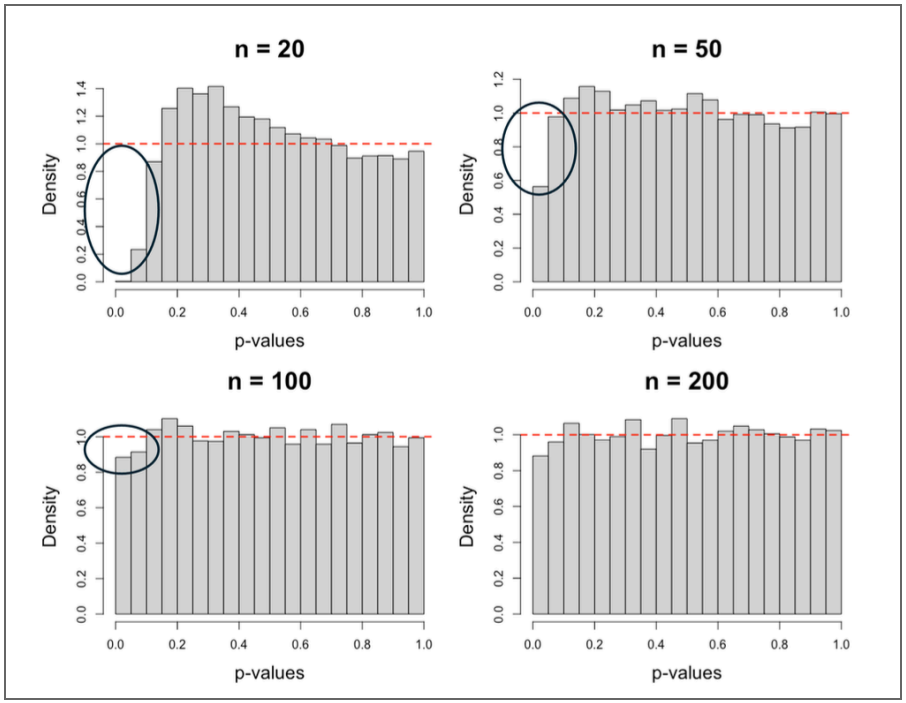
$$T_{\text{Wald}} = \hat{\boldsymbol{\beta}}_{1:p}^T \hat{\Sigma}_{1:p}^{-1} \hat{\boldsymbol{\beta}}_{1:p}.$$

4. $T_{\text{Wald}} \dot{\sim} \chi_p^2$ asymptotically (i.e. as $n \to \infty$) when the null hypothesis is true.

Wald and likelihood ratio tests are asymptotically equivalent under the usual regularity conditions, but can behave differently in small samples.

Hypothesis Testing 2: Wald Test

```r
n <- 20
p_values <- numeric(10000)
for (sim in seq_along(p_values)) {
  x1 <- rnorm(n)
  x2 <- runif(n)
  y <- rbinom(n, 1, 0.3)
  fit <- glm(y ~ x1 + x2,
             family = binomial())
  b <- coef(fit)[2:3]
  V <- vcov(fit)[2:3, 2:3]
  stat <- as.numeric(t(b) %*% solve(V, b))
  p_values[sim] <- pchisq(stat, df = 2,
                          lower.tail = FALSE)
}
```



Checking assumptions

Residuals to check assumptions

In linear regression, residuals help assess assumptions related to ϵ_is

In non-Gaussian GLMs, what specific assumptions are we assessing?

- we don't have such ϵ terms in logistic/Poisson GLMs

The distributional assumptions could be checked indirectly by studying the mean-variance relationship because

- in logistic regression: $Var(y_i) = \mu_i(1 - \mu_i)$
- in Poisson regression: $Var(y_i) = \mu_i$

Still, we need to define a good residual term for GLM so that we can evaluate such assumptions.

Residuals

Pearson ('standardized') residuals:

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{v(\hat{\mu}_i)}}.$$

Deviance residuals:

$$r_i^D = \text{sign}(y_i - \hat{\mu}_i)\sqrt{2\{\ell_i(y_i) - \ell_i(\hat{\mu}_i)\}}.$$

- Here, $\ell_i(y_i)$ is the log likelihood from the saturated model (i.e. a hypothetical model that perfectly fits the data so that $\hat{\mu}_i = y_i$)
- $\ell_i(\hat{\mu}_i)$ is the log likelihood from the fitted model

Residuals: Poisson example

For a fitted Poisson regression,

$$r_i^P = \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}.$$

To calculate the deviance residual, we start with the Poisson log-likelihood for one observation:

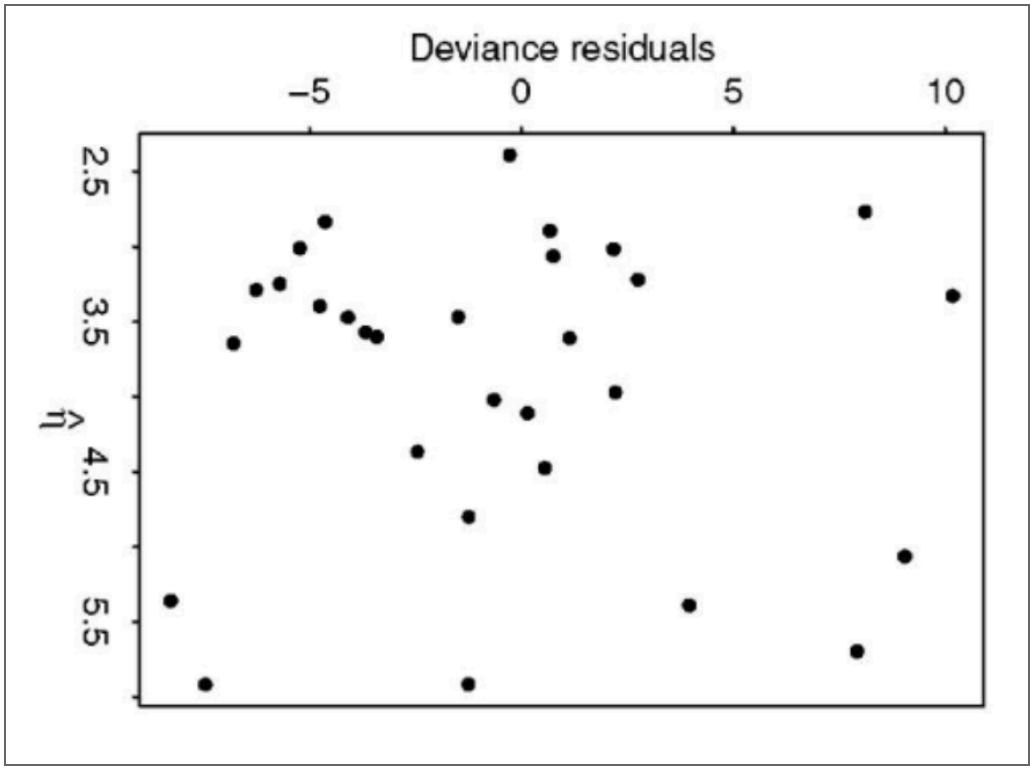
$$\ell_i(\mu_i) = y_i \log(\mu_i) - \mu_i - \log(y_i!).$$

To get the log likelihood of the saturated model, we plug in y_i for μ_i.
To get the log-likelihood of the fitted model, we plug in $\hat{\mu}_i$ for μ_i.
Then, the deviance residual is

$$r_i^D = \text{sign}(y_i - \hat{\mu}_i)\sqrt{2\left[y_i \log\left(\frac{y_i}{\hat{\mu}_i}\right) - (y_i - \hat{\mu}_i)\right]}.$$

Checking ‘Linearity’

We can look at the scatterplot between the deviance residuals and $\mathbf{x}_i^T \hat{\boldsymbol{\beta}}$ to check if there is any non-linear pattern.


Deviance residuals
−5
0
5
10
2.5
3.5
4.5
5.5
η̂

Checking overdispersion and underdispersion

The assumed distribution imposes a mean–variance relationship, so we may perform exploratory analyses to understand overdispersion.

Overdispersion is when the data has more variance than the statistical model expects it to have.

For example, in Poisson, $\hat{\mu}_i$ and $(y_i - \hat{\mu}_i)^2$ characterize the mean and variance. We plot both to compare and look to see if the points are randomly scattered around the 45-degree line.

$(y-\hat{\mu})^2$
0
2
4
6
8
10
2.5
3.5
4.5
5.5
$\hat{\mu}$

Checking Goodness of fit

Two useful summaries are

- Deviance: $D = \sum_{i=1}^{n} (r_i^D)^2$,
- Pearson X^2 statistic: $X^2 = \sum_{i=1}^{n} (r_i^P)^2$

Both D and X^2 follow χ^2_{n-p-1} asymptotically when the mean-variance relationship imposed in your DGP is met.

- If D or X^2 is significantly larger than $n - p - 1$, it suggests a possible overdispersion
- If D or X^2 is significantly less than $n - p - 1$, it suggests a possible underdispersion

Modeling considerations

Modeling Considerations

Recall the motivating question at the start of last lecture: Should we use logistic or Poisson regression just because the data type is binary or count?

Strong distributional assumptions (in variance) specifying expected value and variance

- Bernoulli: $\text{Var}(Y \mid X) = \mu(1-\mu)$.
- Poisson: $\text{Var}(Y \mid X) = \mu$.

Note: When using the normal distribution, there are two separate parameters for the mean and variance.

We may perform exploratory data analysis to check if this assumption is true.

Exposure

Suppose we want to compare the number of infections across two regions.

Region	Population	Number of infections
A	10,000	100
B	1,000	50

Region A has **more infections** than Region B, but Region B has a **higher infection rate**.

The population size determines how much exposure each region has:

> More people $\rightarrow$ more opportunities for infections to occur.

If we model only the raw count, differences in exposure can be mistaken for differences in the underlying event rate.

Exposure

For count data, observations may have different amounts of exposure.
For example:

- number of infections across regions with different population sizes
- number of accidents observed over different driving times
- number of hospitalizations over different follow-up periods

If u_i is the exposure, we can model

$$\log(\mu_i) = \log(u_i) + \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip}.$$

The term $\log(u_i)$ is called an offset.
Equivalently,

$$\log\left(\frac{\mu_i}{u_i}\right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip}.$$

So we are modelling an **event rate per unit exposure**, rather than simply the total count.

Separation

One problem we might face when fitting a logistic regression model is separation. Complete separation occurs when a predictor, or combination of predictors, perfectly separates the two outcome classes.
For example:

- All observations with (x < 5) have (Y=0)
- All observations with (x > 5) have (Y=1)

In this case, the likelihood keeps increasing as the magnitude of the coefficient increases:

$$|\hat{\beta}_1| \to \infty.$$

Therefore, there is **no finite MLE** for

$$\beta_1$$

.

Separation

What might we see in practice?

- Extremely large coefficient estimates
- Extremely large standard errors
- Convergence warnings in R
- Fitted probabilities very close to 0 or 1

What do we do?

- DO NOT just interpret the huge coefficient
- Under separation, the issue is not simply bias: the usual finite maximum likelihood estimate does not exist.
- Use a method that stabilizes the estimate
- penalized logistic regression
- Firth logistic regression

Over/Underdispersion

Quantifying goodness of fit

From the exponential family, it is shown that

$$\mathrm{Var}(y_i) = \phi v(\mu_i) \implies \phi = \frac{\mathrm{Var}(y_i)}{v(\mu)}.$$

where $\phi = 1$ in the Bernoulli or Poisson case.
We may estimate the dispersion parameter, ϕ, by using the X^2 statistic because

$$\frac{X^2}{\phi} \sim \chi^2_{n-p-1} \implies \mathbb{E}\left[\frac{X^2}{n-p-1}\right] = \phi \implies \hat{\phi} = \frac{X^2}{n-p-1}$$

- $\hat{\phi} \gg 1$: suggests strong evidence of overdispersion
- $\hat{\phi} \ll 1$: suggests strong evidence of underdispersion
- $\hat{\phi} \approx 1$: broadly consistent with its variance assumption.

Over/Underdispersion

Quantifying goodness of fit

R Code | Start Over | Run Code

```r
set.seed(0)
n <- 300
x <- rnorm(n)
y <- rnbinom(n, mu = 3, size = 2)
fit <- glm(y ~ x, family = poisson())
sum(residuals(fit, type = "pearson")^2) / df.residual(fit)

set.seed(0)
x <- rnorm(n)
y <- rpois(n, lambda = 3)
fit <- glm(y ~ x, family = poisson())
sum(residuals(fit, type = "pearson")^2) / df.residual(fit)
```

```
[1] 2.518681
```

```
[1] 1.039035
```

Dispersion: impact on inference

With overdispersion:

- The actual variance of data is greater than your modeled (assumed) variance
- It means that the your model underestimates the true variance of parameter estimates
- Your test statistic takes variance of parameter estimates as a denominator, which implies that the test statistic will be greater than what it should be
- It eventually leads to inflation in Type 1 error.

Similarly, your Type 1 error would be overly conservative when underdispersion occurs.

- But it is ‘okay’ in preventing inflated T1E.
- That’s one of the reasons that underdispersion is considered less serious than overdispersion

Overdispersion: remedies

1. Use a more flexible DGP (i.e. where mean and variance are specified by two parameters).
   - Grouped binary counts: beta-binomial distribution.
   - Counts: negative binomial distribution, for example `MASS::glm.nb()`.
2. Use quasi-likelihood with an adjusted variance.
   - The variance of $\hat{\boldsymbol{\beta}}$ is $\boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X}$, with $\phi = 1$ in logistic and Poisson GLM
   - We know that $\hat{\phi} > 1$ when overdispersion occurs
   - Key Idea: the variance of $\hat{\boldsymbol{\beta}}$ is replaced by $\hat{\phi} \cdot (\boldsymbol{X}^T\boldsymbol{W}\boldsymbol{X})$ with $\hat{\phi}$ estimated from the GLM fit (Pearson $X^2 statistic$)
   - Does not change any $\hat{\boldsymbol{\beta}}$, but provides a more robust variance estimate.
   - Aims to prevent inflated T1E

In R: `quasipoisson(link = "log")` or `quasibinomial(link = "logit")` in `glm()` to implement Quasi-likelihood.

More on quasi-likelihood

- Quasi-likelihood does not characterize any specific data-generating process.

- Therefore, you cannot use Likelihood Ratio Test with quasi-poisson or quasi-binomial models (because there is no such likelihood).

- Welcome to robust inference!
- M-estimation
- Sandwich covariance estimators
- Generalized estimating equations (GEE)
- Generalized method of moments (GMM)

Lars Peter Hansen received the 2013 Nobel Prize in Economics for contributions including GMM.

Summary

- Generalized linear models extend linear regression to responses such as binary and count data by specifying a response distribution, a linear predictor, and a link function.
- GLM coefficients are typically estimated by maximum likelihood, often using an iterative procedure such as iteratively reweighted least squares.
- Statistical inference in GLMs relies largely on asymptotic results; Wald tests and likelihood ratio tests can be used to assess covariate effects.
- Model checking requires assessing both the mean structure and the distributional mean–variance relationship using residuals and goodness-of-fit measures.
- When assumptions such as the Poisson variance relationship are violated, alternatives such as negative binomial models, zero-inflated models, or quasi-likelihood may provide more appropriate modelling or inference.

Announcements: Next week is the midterm (Friday, October 23rd) in room FE 230 from 2:10pm – 4:10pm

Speaker notes